

YASH PURSWANI

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Google Scholar

Education

Indian Institute of Technology Madras

Jul 2022 - Jul 2027

B.Tech. in Mechanical Engineering and Inter-Disciplinary M.Tech. in Data Science

9.24/10.00

Research Interests

Robot Perception · Multi-Robot Control · Reinforcement Learning · Autonomous Systems

Publications

- Rule-Constrained Adversarial Scenario Generation, *IEEE Intelligent Vehicles Symposium (IV)*, Manuscript in preparation
- Robust Finite-Time Control of Multi-Agent Swarms, *IEEE American Control Conference (ACC)*, 2027, Manuscript in preparation

Research and Project Experience

[TITLE for MS THESIS]

DAAD KOSPIE Scholar, Prof. Aamir Ahmad, IFR, Universität Stuttgart

(Sept 2026 - Present)

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Rule-Constrained Adversarial Scenario Generation

RobotX 2026 Fellow, Prof. Emilio Frazzoli, IDSC, ETH Zürich

(Jul 2026 - Aug 2026)

- Proposed a novel RL framework for generating **critical** yet **realistic adversarial scenarios** for AV safety validation
- Reformulated adversarial scenario generation with **rule-constrained non-ego** agents using feasibility-aware learning
- Demonstrated that **feasibility learning** can serve as a **reliable surrogate** for learning and enforcing rules and constraints
- Extended the **FREA** framework with feasibility-guided RL, reducing rule violations while preserving adversarial behaviour

Robust Finite-Time Control of Multi-Agent Swarms

[Project Link](#)

Guide: Prof. Anuj Kumar Tiwari, ME Dept., IIT Madras

(Sept 2025 - Sept 2026)

- Proposed a hybrid swarm kinematic model enabling **finite-time** centroid tracking independent of formation geometry
- Proved global exponential convergence of formation errors using distance-based inter-agent coupling
- Established **Input-to-State Stability (ISS)** under bounded disturbances with analytically characterized invariant error bound
- Derived **super-linear disturbance attenuation** for centroid tracking using finite-time sliding-mode control

Multi-View 3D Scene Reconstruction Pipeline for Robotic Perception

[Project Link](#)

Guide: Dr. Sudarsun Santhiappan, DSAI Dept., IIT Madras

(Mar 2026 - May 2026)

- Built a **multi-view 3D reconstruction** pipeline from unordered images using **SfM**, feature matching and dense reconstruction
- Estimated **camera trajectories** and dense point clouds using multi-view geometry for robotic scene understanding
- Developed an **end-to-end scalable** pipeline for reconstructing real-world 3D environments and containerized deployment

Autonomous Road Dust Collector

Guide: Prof. Sivanagendra SM, CE Dept., IIT Madras

(Nov 2025 - Mar 2026)

- Developed an **Autonomous Road Dust Collection Robot** for unmanned operations in sub-urban environments
- Implemented 3D LiDAR-based odometry, and mapping using **FAST-LIO2** and **NDT-based localization** for pose estimation
- Incorporated **Lanelet-based** semantic road modelling for path planning and enable context-aware autonomous navigation
- Performed system **performance evaluation** and **failure-mode analysis** using time-series sensor data and simulation logs

Optimal Rendezvous Trajectory for Unmanned Aerial-Ground Vehicles

Guide: Prof. Rachel Kalpana Kalaimani, EE Dept., IIT Madras

(Mar 2025 - May 2025)

- Designed an **optimal control** framework for UAV-UGV rendezvous and **in-motion refuelling** under non-linear dynamics
- Developed a trajectory-tracking and **aggressiveness-index-based** control strategy enabling **tunable rendezvous** behaviour
- Validated the method through numerical simulations in PRONTO, demonstrating robust convergence and adaptability

Professional Experience

Team Abhiyaan — IIT Madras (Apr 2023 - May 2026)

Team Lead / Software Module Lead / Robotics Engineer

BOLT — Driverless Institute Shuttle

Guide: Prof. Sathyan Subbiah, ME Dept., IIT Madras

- Developed and fully deployed a **ROS-based software** on an electric golf cart, turning it into a **Driverless Institute Shuttle**
- Achieved **98.3% accuracy** using 3D LiDAR-camera sensor fusion for precise object classification and distance estimation
- Leveraged realistic **CARLA** and **Gazebo** simulations to conduct comprehensive testing of the software subsystem for BOLT

Project Vidyut

Intelligent Ground Vehicle Competition (IGVC), Michigan, USA

- Developed a **ROS2-based navigation** stack integrating **Nav2**, SLAM, and sensor fusion for autonomous navigation
- Implemented ORB-SLAM3, SLAM Toolbox, CUDA-accelerated perception, and EKF-based sensor fusion for localization
- Led software development and system integration for reliable autonomous vehicle deployment at IGVC 2024
- Secured **6th position** in the Design Challenge and an overall **7th position** among 30+ international teams at IGVC 2024

Team Leadership

- Led a **30+** member team, overseeing software architecture, development, and validation of autonomous robots

Honors & Awards

- RobotX 2026 Fellow, ETH Zürich, selected for competitive robotics research fellowship
- DAAD KOSPIE 2026 Scholar, selected for master's thesis exchange at Universität Stuttgart
- IGVC 2024, 7th Overall among 30+ international teams and 6th in Design Challenge
- Inter-IIT Tech Meet 2024, Silver Medal for designing an efficient air-conditioning system

Technical Skills

- **Programming Languages:** Python, C++, MATLAB, Bash
- **Robotics:** ROS2, Gazebo, CARLA, NVIDIA Omniverse
- **Perception:** OpenCV, Open3D, FAST-LIO2, FAST-LIVO2, ORB-SLAM3
- **ML/RL Libraries:** Pytorch, Tensorflow, scikit-learn, Gymnasium
- **Systems:** Linux, CUDA, CMake, Docker, Git

Relevant Courses

- **Robotics, Dynamics & Control:** Network Dynamics & Controls · Optimal Control · Control Systems · Motion Planning for Self-Driving Cars*
- **Learning:** Reinforcement Learning · Foundations of Machine Learning · Machine Learning Operations
- **Others:** Parallel Scientific Computing · Multi-Variable Calculus · Linear Algebra · Data Structures & Algorithms